Code Availability Validation Report v1: “A Unified Database of the Central Economic Mathematical Institute of the USSR Academy of Sciences, 1961–1987”

GitHub Repository and Live End-to-End Reproduction Audit

Date of audit: 2026-07-07 Report version: v1 Auditor: Claude (Anthropic AI assistant) Files and resources examined: - Paper: A Unified Database of the Central Economic Mathematical Institute of the USSR Academy of Sciences, 1961–1987.docx - Manual: README.md / README.txt / README.docx (repository HEAD, commit 57e0d0f) - Repository: https://github.com/SovietEconLab/CEMI-Unified-Database (cloned at HEAD)

Scope note: per the authors’ archiving plan, long-term preservation is handled automatically through the GitHub–Zenodo integration under the concept DOI 10.5281/zenodo.21254566. Every archived artefact is generated from the GitHub repository, so this audit validates the repository itself.

Executive Summary

A **live end-to-end reproduction** of the deposit was performed in a clean environment (Ubuntu 22.04.5, Python 3.10.12, SQLite 3.37.2 — deliberately different from the development stack, Python 3.11 / SQLite 3.50.4), following the deposit README literally.

Every documented count was reproduced exactly, at both build tiers. The lean build yielded 41 tables · 55,711 rows · 1,954 persons · 12,211 person-year observations · 181 sheets_index rows · 636 classified institutions; the full-fidelity build (README §B.1 Tier 2) yielded 56,931 rows · 757 sheets_index rows · 652 classified institutions — all matching the paper, README, and codebook to the row. Two independent lean builds were **bit-identical** (SHA-256 verified), both tiers passed `PRAGMA integrity_check` and `foreign_key_check` with zero violations, and the interface generator produced the documented ~7.5 MB self-contained HTML with all twelve documented tabs. The prebuilt database shipped in folder 05. matches the lean-tier counts exactly and was written by SQLite 3.50.4, as the paper states.

The repository tree matches the README Section 0 inventory exactly, the paper title cited in the README matches the manuscript, the concept DOI is cited uniformly across paper and README (12 occurrences in the paper; no stale identifiers), and the licence declaration (CC BY 4.0 for the entire deposit) is consistent throughout.

Overall verdict: PASS — fully reproducible as documented; no discrepancies found.

1. Scope and Method

The audit executed the reproduction protocol of the paper's Technical Validation section from scratch: clone the public repository, follow the deposit README literally, rebuild the database at both tiers in clean working directories, rebuild the interface, and compare every observable count against the claims in the paper, README, and codebook.

Validation environment:

Component	Development stack (paper)	Paper's validation env	This audit
OS	(not stated)	Ubuntu 22.04	Ubuntu 22.04.5 LTS
Python	3.11	3.10.12	3.10.12
SQLite	3.50.4	3.37.2	3.37.2
pandas / openpyxl	(unpinned)	(not stated)	2.3.3 / 3.1.5

2. GitHub Repository Status

Check	Result
`git clone https://github.com/SovietEconLab/CEMI-Unified-Database`	✅ Succeeded (public); HEAD = 57e0d0f
Working tree	44 MB; folders `01.`–`05.` all present (01: 4 sample PDFs + 2 script copies; 02: 5 files; 03: 25 workbooks; 04: 25 + 3 auxiliary workbooks + glossary(0530).xlsx; 05: 7 files incl. prebuilt artefacts)
Root files vs README Section 0 inventory	✅ Matches exactly (README ×3, two institution glossaries, ARAN document lists, two codebooks)
Paper title in README	✅ Matches the manuscript title
Licence statement (README ↔ paper)	✅ CC BY 4.0 for everything in the repository — consistent
Concept-DOI citation policy (README ↔ paper)	✅ Consistent — concept DOI 10.5281/zenodo.21254566 cited uniformly; no stale identifiers

3. Live Reproduction — Lean Tier (README §B.1 Tier 1)

Procedure followed literally: the builder, interface generator and the three Excel inputs were copied into two clean working directories with no extra files co-located; `python cemi_career_db.py` was run in each.

Claim (paper & README)	Value stated	Observed	Match?
Build exit status	0	0 (`[OK] Built cemi_career.db`)	✅
Tables	41	41	✅
Total rows	55,711	55,711	✅
Persons identified	1,954	1,954	✅
Person-year observations	12,211	12,211	✅
sheets_index rows	181	181	✅
Classified institutions	636	636 (Academy 83 + Branch 335 + Enterprise 35 + VUZ 108 + Other 75)	✅
Nine source event-fact tables, total rows	19,708	19,708	✅
Per-year observations range	21 (1964) – 1,084 (1979)	21 (1964) – 1,084 (1979)	✅
data.xlsx dimensions	12,211 × 27	12,211 × 27	✅
`PRAGMA integrity_check` / `foreign_key_check`	ok / empty	ok / 0 violations	✅
Prebuilt DB in folder 05. matches lean counts	yes	yes (all counts identical)	✅
Prebuilt DB written by SQLite	3.50.4	3.50.4 (file-header check)	✅
Build time	< 5 minutes	< 1 minute	✅
DB size	~5.4 MB	5,361,664 bytes	✅

4. Live Reproduction — Full-Fidelity Tier (README §B.1 Tier 2)

Procedure followed literally: the institution glossary and the two provenance folders (`OCR Results/`, `English Translation Results/`) were co-located next to `data.xlsx` exactly as instructed, then the builder was run.

Claim	Value stated	Observed	Match?
Total rows	56,931	56,931	✅
sheets_index rows	757	757	✅

Classified institutions	652	652 (Academy 85 + Branch 342 + Enterprise 38 + VUZ 111 + Other 76)	✅
Persons / person-year observations	1,954 / 12,211	1,954 / 12,211	✅
Tables	41	41	✅
`PRAGMA integrity_check` / `foreign_key_check`	ok / empty	ok / 0 violations	✅

The full−lean delta decomposes exactly as documented: +628 glossary rows, +16 institutions, +576 sheets_index provenance rows (147 archival + 34 workbook rows in both tiers; the folder scan adds 282 OCR-intermediate and 294 English-translation rows).

5. Interface Generation

`python cemi_career_ui.py` completed with exit 0 and produced `cemi_career_ui.html` of 7,520,597 bytes (~7.5 MB claimed ✅) using only the Python standard library, as documented. All twelve documented tabs are present in the generated file: Overview, Career search, Institutions, Positions, Personnel growth, Degrees & titles, Nationality, Age distribution, Research Fields, Party & trainees, Provenance, Glossary. ✅

6. Determinism and Hash Behaviour

- **Lean tier**: two independent builds in separate clean directories were **bit-identical** (SHA-256 `3a069796`…). ✅
- **Full-fidelity tier**: repeated builds on identical inputs were **bit-identical**. ✅
- The locally rebuilt lean DB differs byte-wise from the prebuilt DB (SHA-256 `ef1d0f5d`…) while agreeing on every count — exactly the cross-SQLite-version behaviour (3.37.2 vs 3.50.4) documented in the paper's Technical Validation. ✅

7. Issues and Discrepancies Found

None. Every quantitative claim checked in the paper, README, and codebook was reproduced exactly. Two cosmetic observations, both already documented by the authors and requiring no action: the deposit provides `requirements.txt` inside `02. OCR and Translate Code/` rather than at the repository root (the README states this explicitly), and the README's verification snippet uses the Unix `shasum` command (Windows users can substitute `Get-FileHash`; the primary cross-platform verification path is the Python snippet).

8. Reproducibility Test Outcome

What was tested (live): ✅ public clone at current HEAD — ✅ deposit structure vs README Section 0 (exact) — ✅ lean build, twice, in clean directories (all 16 checked claims exact)

— ✅ full-fidelity build per §B.1 Tier 2 (all checked claims exact) — ✅ delta decomposition between tiers (exact) — ✅ determinism at both tiers (SHA-256) — ✅ integrity and foreign-key checks at both tiers — ✅ interface generation, size, and tab inventory — ✅ prebuilt-artefact consistency and writing-library version.

9. Summary Verdict

Category	Status
Repository accessibility	✅ PUBLIC — clonable; inventory matches README exactly
Lean-tier reproduction	✅ EXACT — every documented count reproduced, bit-stable
Full-fidelity reproduction	✅ EXACT — every documented count reproduced
Interface	✅ REPRODUCED — size and twelve tabs as documented
Determinism claims	✅ CONFIRMED — bit-identical rebuilds; cross-version hash caveat behaves as documented
Paper ↔ README ↔ codebook consistency	✅ CONSISTENT — title, DOI policy, licence, and every substantive count aligned

Overall Assessment

The deposit is **fully reproducible**: an independent, from-scratch rebuild in an environment different from the development stack, following the README literally, reproduced every documented count exactly at both build tiers, deterministically, with clean integrity checks. The paper, README, codebook, and repository are mutually consistent on the title, licence, citation policy, and all quantitative claims. The documentation is detailed enough that no step required interpretation or correction. No action items result from this audit.

10. Recommendations

1. None required for reproducibility. As routine hygiene for future releases: after each new tagged release, download the released archive once and verify its checksum as a packaging check.